

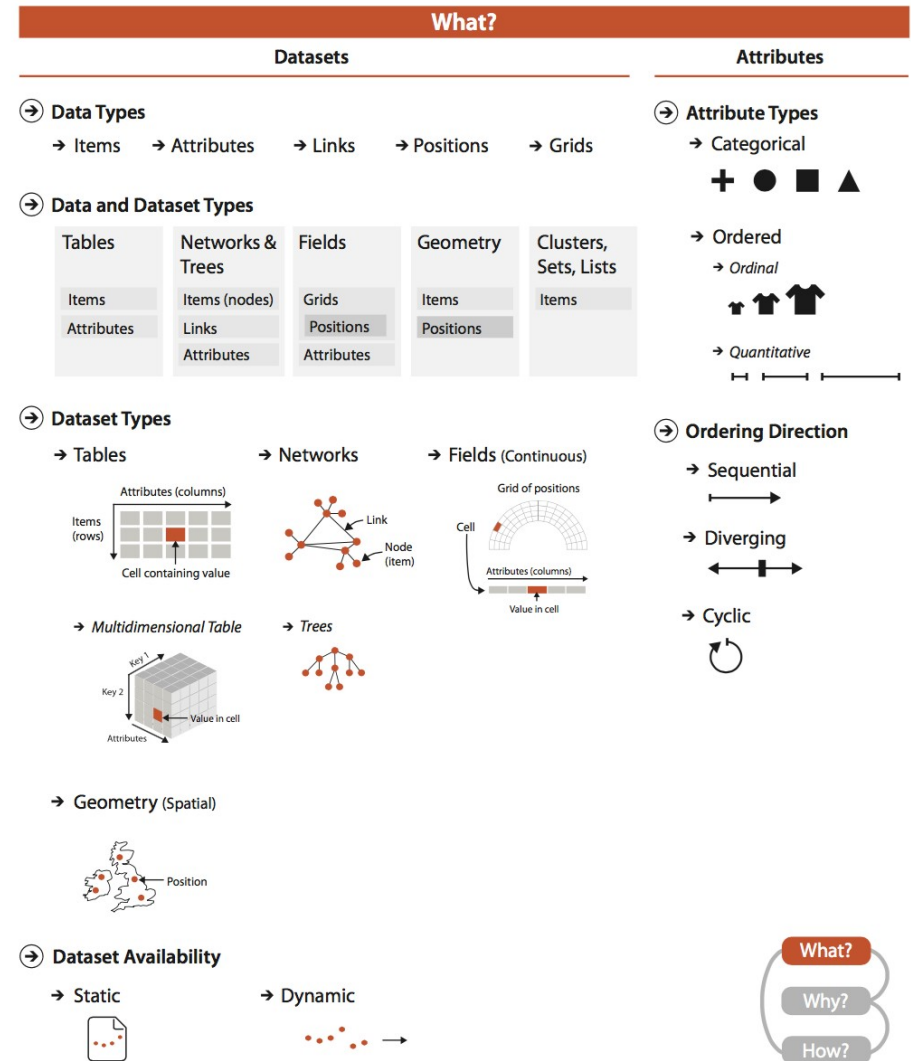
Data Abstraction and Representation

Denis L. Nkweteyim

Outline

- Data Types for Visualization
- Dataset Types

Data Types and Dataset Types



Data Types

- Munzner defines 5 fundamental data types
 - Items
 - Attributes
 - Links
 - Positions
 - Grids

Data Types

- Item
 - A discrete individual entity - such as a row in a tidy table, or a node in a network
- Attributes
 - Also called variables or data dimensions
 - Specific property that can be measured, observed, or logged

Data Types - Attributes

- Attributes types
 - Categorical Data
 - Does not intrinsically support arithmetic operations or a total ordering
 - Categorical (nominal)
 - No total ordering
 - Examples
 - Names (of brands, people, places, ...)
 - Movie genres
 - File types

Data Types – Attribute

- Attribute types
 - Ordered Data
 - Data that has an implicit (total) ordering
 - Two types
 - Ordered (categorical)
 - Also known as ordinal
 - Does not support arithmetic operations
 - Examples: shirt size (ordinal), rankings (ordinal)
 - Quantitative
 - Data that can be measured or counted
 - Supports arithmetic operations
 - Examples: weight, height

Data Types – Attribute

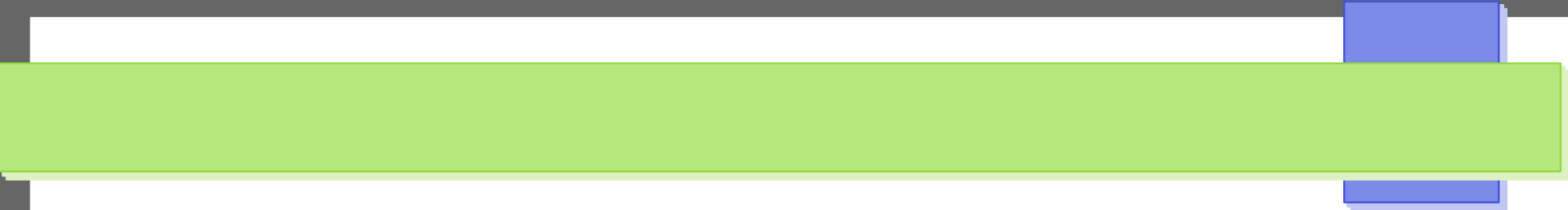
- Types of quantitative data
 - Integers vs reals
 - Interval data vs ratio data
 - Interval data
 - The difference between values is meaningful
 - May not have a meaningful 0-value
 - Ratio data
 - The ratio between values is meaningful
 - May not have a meaningful 0-value

Data Types – Attribute

- Interval data
 - Provides ordered, ranked values with equal spacing between them, but the zero point is not absolute
 - Can measure the distance between points, but cannot calculate ratios
 - e.g., 20 °C is not "twice as hot" as 10°C
 - Examples: Temperature (Celsius/Fahrenheit), IQ scores, Calendar years (2024, 2025)

Data Types – Attribute

- Ratio data
 - The most precise level of measurement, possessing all properties of interval data but with a meaningful zero point
 - 0 represents the total absence of the quantity
 - Ratios are meaningful (e.g. 20kg is exactly twice as heavy as 10kg)
 - Examples: Weight, height, time duration, distance, income.

- 
- Ordered data – Ordering direction
 - Sequential
 - Diverging
 - Cyclic
 -

- Sequential

- Goes from a minimum value to a maximum value
- Does not have to have a meaningful basepoint or 0-value.
- Examples
 - Height or weight of a person, course grades, taxation brackets

Data Types

- Diverging ordinal data
 - Emerges from one neutral center in two (or, rarely, more) directions of progressively more extreme values.
 - Examples
 - Temperature (above/below freezing)
 - Altitude (above/below sea level)

Data Types

- Cyclic
 - An attribute is cyclic if its values wrap around to a starting point after a while
 - Examples
 - Time of day, day of week, day of year, year of century
 - Also compass directions

Data Type Links

- Connections between items, usually in networks and graphs
 - A link will connect two items to each other
 - Examples
 - Friends/follows on social networks
 - Connectivity of roads or along train lines
 - Family relationships
 - Organizational charts

Data Type Positions

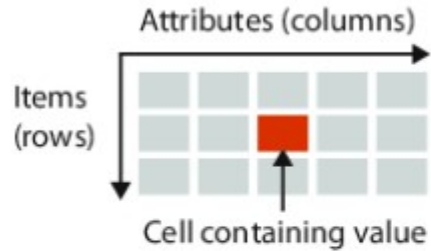
- Position data corresponds to locations in spatial data - primarily as 2-dimensional or 3-dimensional points
- Location in space, typically given as a set of coordinates
- Examples
 - Latitude and longitude
 - x and y coordinates

Data Type Grids

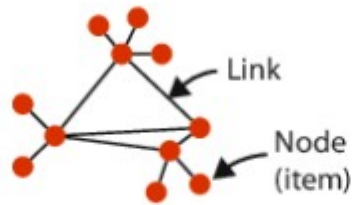
- Grid acts as a structured framework, defined by its geometry and topology, that connects positions in space to data
 - Examples: square grid, triangular grid, hexagonal grid
- Structure
 - Grids can be uniform, regular, or irregular, and they are typically used to represent spatial data where each cell contains one or more data attributes
- Scalar/Vector Fields
 - Grids often contain
 - Scalar data (a single quantitative attribute per grid cell, like temperature) or
 - Vector/tensor data (multiple attributes per cell)

Types of Datasets

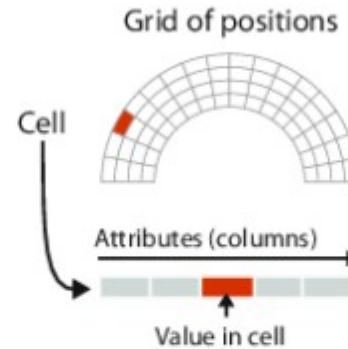
→ Tables



→ Networks



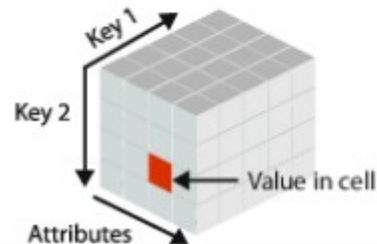
→ Fields (Continuous)



→ Geometry (Spatial)



→ Multidimensional Table



→ Trees



Types of Datasets

- Core dataset types
 - Tables
 - Structured as rows (items) and columns (attributes)
 - Each variable must have its own column
 - Each item (observation) must have its own row
 - Each value must have its own cell
 - Note
 - Tabular (cell) data also includes multidimensional tables (data hypercubes; tensors) that have composite keys

Types of Datasets

- Core dataset types
 - Networks & Trees
 - Represent relationships between items
 - Networks consist of nodes and links
 - Trees: A specific type of network without cycles, showing hierarchy
 - Fields
 - Represent continuous, spatial data, often defined on a grid
 - They describe data that exists across a continuous space (e.g., fluid flow, temperature map)
 - Geometry
 - Data that describes the shape of objects, focusing on spatial positions

Types of Datasets

- Other Data Classifications
 - Clusters, Sets, Lists
 - Collections of items without the rigid structure of a table
 - Static vs. Dynamic
 - Static data: Entire dataset available at once
 - Dynamic data: Data trickles in over time, such as streams